

Ankit Gahlawat

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Professional Summary

Passionate and curious Computer Vision Researcher pursuing a PhD with a strong background in software engineering. My research focuses on human face analysis and synthesis driving advancements in multimodal and generative AI for creative applications.

Education

International Institute of Information Technology Bangalore

PhD in Artificial Intelligence (CGPA: 3.8/4), Research Focus: 3D Vision

2024–Present

Advisor: Prof. Dinesh Babu Jayagopi

International Institute of Information Technology Bangalore

Bachelor + Master in Computer Science and Engineering

2015–2020

Research Experience

IIIT Bangalore

PhD Researcher

Multimodal Perception Lab

Aug 2024–Present

- Improving face parsing using 3D Gaussian Splatting (Paper accepted in VCIP 2025).
- Finding best viewing angles for constructing a 3D human face using Bayesian Optimisation.

Openstream

Research Associate (IIIT Bangalore)

Multimodal Perception Lab

Aug 2023–July 2024

- Worked on a project for conversational avatars in real-time settings.
- Researched and implemented the best solutions for audio-driven talking avatar primarily using NeRF and 3D Gaussian Splatting.

Software Engineering Experience

Gojek

Software Engineer - 2

June 2022–July 2023

- Worked in payments and monetisation team in transport BU.
- Launched trip insurance functionality in Singapore and Vietnam.

Dhani

Senior Software Developer

Nov 2021–May 2022

- Worked in core-finance backend team.
- Setup a new Spring Batch service to run all the batch jobs needed for Dhani wallet functionalities.

Flipkart

Software Development Engineer - 1

June 2020–Nov 2021

- As a part of seller growth team, generated recommendations for sellers to help their business grow.
- Built feedback system and seller metrics tools that empower sellers to have deep insights into how their business is performing.

Goldman Sachs

Summer Analyst

May 2019–June 2019

- Worked in Finance Engineering team.
- Built an application to accelerate the Feed Setup process from 1 day to 3 minutes.

Publications

Ankit Gahlawat, Anirban Mukherjee, Dinesh Babu Jayagopi : Efficient Label Refinement for Face Parsing Under Extreme Poses Using 3D Gaussian Splatting

VCIP 2025

Technical Skills

Programming: Python, C++, Java

3D Vision: NeRF, 3D Gaussian Splatting, COLMAP, Structure from Motion (SfM)

Libraries & Tools: OpenCV, Keras, NumPy, SciPy, Matplotlib, SIFT, RANSAC, NeRFStudio

Frameworks: PyTorch, TensorFlow

Tools: Git, Docker

Soft Skills: Complex Problem Solving, Coordinating with Others, Decision Making

Projects

Optimisation

View Planning using Bayesian Optimisation

Developing a view planning model to select optimal camera angles for 3D Gaussian Splatting (3DGS) using Bayesian Optimization. The method automatically identifies the most informative views to minimize training time and computational cost while preserving reconstruction quality.

3D Vision

Camera Calibration and Homography Estimation

Conducted camera calibration using a Rubik's cube as a non-planar object and estimated homography between chessboard images using manual and feature-based matching (SIFT).

3D Vision

Rendering Comparison using NeRF and 3DGS

Implemented 3D rendering of an object using Neural Radiance Fields (NeRF) and 3D Gaussian Splatting (3DGS). The project involved generating a 3D point cloud using Structure from Motion (SfM), and comparing both techniques based on quality, training time, and rendering speed.

Computer Vision

Face Mask Detection, Classification and Segmentation

Built a CV pipeline to classify and segment face masks using both classical ML and DL techniques. Used HOG with SVM/MLP for binary classification and a custom CNN (95% accuracy) with handcrafted image preprocessing. Applied Canny Edge Detection for traditional segmentation (IoU: 0.34) and trained a U-Net model achieving 0.92 IoU for precise segmentation on MFSD dataset.